**3** A thin metal wire X, of diameter $1.2 \times 10^{-3}$ m, is used to suspend a model planet, as shown in Fig. 3.1.

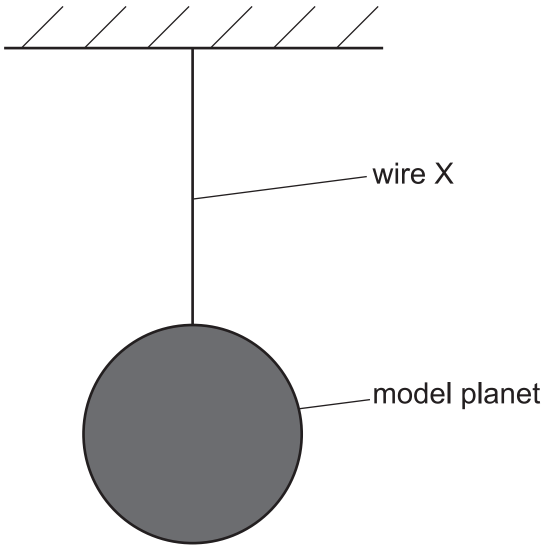



**Fig. 3.1** (not to scale)

The variation with strain of the stress for wire X is shown in Fig. 3.2.

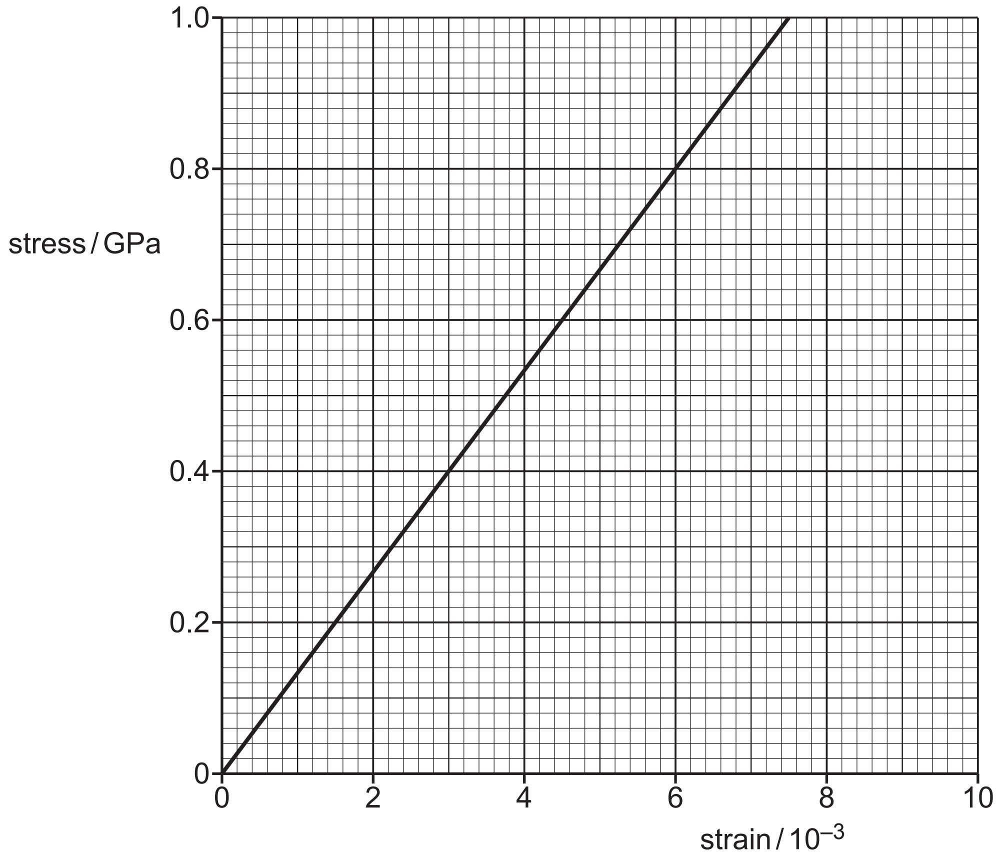



**Fig. 3.2**

**(a)** The strain in X is $5.4 \times 10^{-3}$.

**(i)** Use Fig. 3.2 to calculate the force exerted on the wire by the model planet.

| | | |
|---|---|---|
| $\sigma = 0.72 \times 10^9$ | **C1** | Allow any power of 10 for the first C1 mark<br>Allow 0.71 to 0.73 as tolerance<br>Treat misreads e.g. 0.74 as TE and Allow ECF |
| force = $\sigma \times A$<br>$= 0.72 \times 10^9 \times \pi \times (1.2 \times 10^{-3}/2)^2$ | **C1** | Allow $\pi r^2 = A$ |
| $= 810$ N | **A1** | Allow answers between 800 N and 830 N |
| OR | | |
| Young modulus = gradient of graph<br>e.g. $= 0.80 \times 10^9 / 6.0 \times 10^{-3}$<br>$= 1.33 \times 10^{11}$ | **(C1)** | Allow 'E' for young modulus<br>Young modulus between $1.31 \times 10^{11}$ and $1.35 \times 10^{11}$ as tolerance from graph. |
| force = Young modulus × strain × $A$<br>$= 1.33 \times 10^{11} \times 5.4 \times 10^{-3} \times \pi \times (1.2 \times 10^{-3}/2)^2$ | **(C1)** | |
| $= 810$ N | **(A1)** | Allow answers between 800 N and 830 N |

force = .................................................... N [3]

**(ii)** The elastic potential energy of X is 0.31 J.

Calculate the original length of the wire before the model planet was attached.

| | | |
|---|---|---|
| $E_{(P)} = ½ Fx$<br>or $E_{(P)} = ½ kx^2$ <u>and</u> $F = kx$ | **C1** | Any subject<br>Allow e or ΔL for x |
| $x = 2E_P / F$<br>$x = 2 \times 0.31 / 810$<br>$= 7.7 \times 10^{-4}$ | **C1** | ECF Force from 3(a)(i)<br>($x = 7.75 \times 10^{-4}$ m if 800N carried forward from (a)(i))<br>($x = 7.47 \times 10^{-4}$ m if 830N carried forward from (a)(i)) |
| $L = x / \varepsilon$<br>$L = 7.7 \times 10^{-4} / 5.4 \times 10^{-3}$<br>$L = 0.14$ m | **A1** | |
| **OR** | | |
| $E_{(P)} = ½ Fx$<br>or $E_{(P)} = ½ kx^2$ <u>and</u> $k = EA/L$ | **(C1)** | (and $F = EAx/L$) |
| $x = 2E_P / EA\varepsilon$<br>$x = 2 \times 0.31 / (1.33 \times 10^{11} \times \pi \times (1.2 \times 10^{-3}/2)^2 \times 5.4 \times 10^{-3})$<br>$x = 7.6 \times 10^{-4}$ | **(C1)** | ECF Young Modulus from 3(a)(i)<br>(x = 7.52 if $1.35 \times 10^{11}$ used for E)<br>(x = 7.75 if $1.31 \times 10^{11}$ used for E) |
| $L = x / \varepsilon$<br>$L = 7.6 \times 10^{-4} / 5.4 \times 10^{-3}$<br>$L = 0.14$ m | **(A1)** | |

[3]

**(b)** Wire X is replaced by a new wire, Y, with the same original length and diameter but double the Young modulus of X. Wire Y also obeys Hooke's law.

On Fig. 3.2, draw a line representing the variation with strain of the stress for Y. [2]

[Total: 8]

| | | |
|---|---|---|
| A straight line, passing through the origin with a larger gradient than wire X | **M1** | line must go from origin (tolerance of 1 small square vertically and horizontally) to at least 0.5 GPa<br>Allow free-hand straight line. Ruler from origin to furthest point on line should not give any deviation of more than 1 square horizontally. |
| Gradient of the line is twice the gradient of wire X | **A1** | line passing through (3, 0.8) +- 1 small square horizontally (extrapolate with ruler tool if necessary) |

**4** **(a)** Define strain.

| | |
|---|---|
| extension / <u>original</u> length | **B1** |

.......................................................................................................................................... [1]

**(b)** A copper wire of length 4.0 m has a uniform cross-sectional area of $4.5 \times 10^{-7}\,\text{m}^2$.

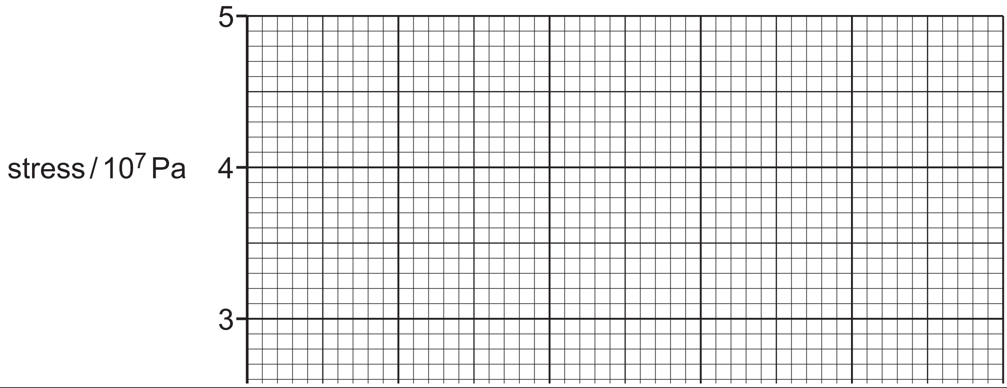
A tensile force of 18 N is applied to the wire. This causes the wire to extend by 1.4 mm up to its limit of proportionality.

**(i)** Calculate the Young modulus of the wire.

| | | |
|---|---|---|
| Young modulus | = stress / strain | **C1** |
| | $= (18 / 4.5 \times 10^{-7}) / (1.4 \times 10^{-3} / 4.0)$ | **C1** |
| | $= (4.0 \times 10^{7}) / (3.5 \times 10^{-4})$<br>$= 1.1 \times 10^{11}$ Pa | **A1** |

Young modulus = ..................................................... Pa [3]

**(ii)** On Fig. 4.1, draw a line to show how the stress varies with the strain for the wire up to its limit of proportionality.




| | |
|---|---|
| straight line through the origin | **B1** |
| ending at the point (3.5, 4.0) | **B1** |

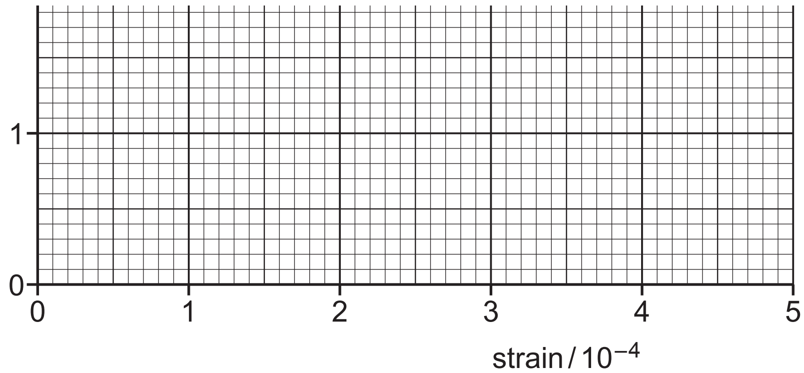



**Fig. 4.1**

[2]

**(c)** A second copper wire has the same length as the wire in **(b)** but a larger diameter. Both wires are subjected to a tensile force of 18 N.

By placing a tick (✓) in each row, complete Table 4.1 to compare the stress and strain of the two wires.

**Table 4.1**

| | greater in second wire | less in second wire | the same in both wires |
|---|---|---|---|
| stress | | | |
| strain | | | |

[2]

| | greater in second wire | less in second wire | the same in both wires |
|---|---|---|---|
| stress | | ✓ | |
| strain | | ✓ | |

**B2**

[Total: 8]

**3** **(a)** State Hooke's law.

| | |
|---|---|
| extension is proportional to (applied) force | **B1** |

............................................................................................................................................ [1]

**(b)** The variation of the applied force with the extension for a sample of a material is shown in Fig. 3.1.

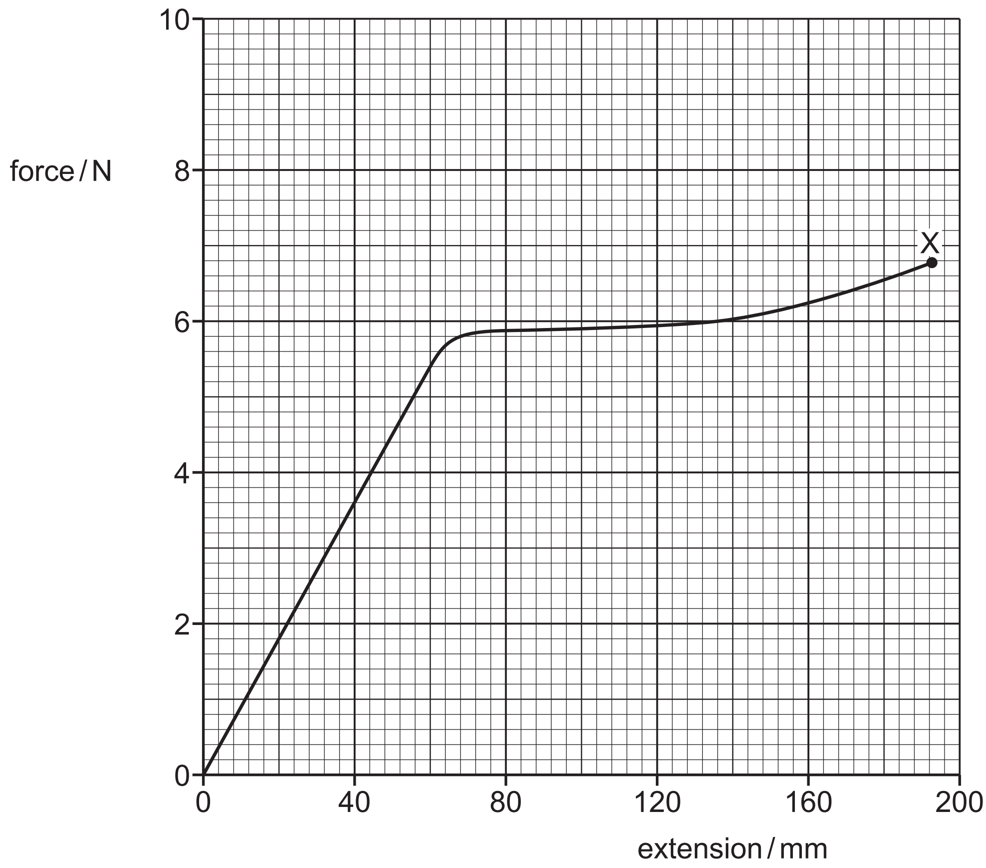



**Fig. 3.1**

The sample behaves elastically up to an extension of 80 mm and breaks at point X.

**(i)** On the line in Fig. 3.1, draw a cross (×) to show the limit of proportionality. Label this cross with the letter P. [1]

**(ii)** On the line in Fig. 3.1, draw a cross (×) to show the elastic limit. Label this cross with the letter E. [1]

| | |
|---|---|
| P at (60, 5.4) | **A1** |
| E at (80, 5.9) | **A1** |

**(c)** The sample in **(b)** has a cross-sectional area of $0.40\,\text{mm}^2$ and an initial length of 3.2 m.

For deformations within the limit of proportionality of the sample, determine:

**(i)** the spring constant of the sample

| | |
|---|---|
| $k = F / x$ **or** $k$ = gradient of (straight line section of) graph | **C1** |
| e.g. gradient = 5.4 / 0.060<br>$k = 90\,\text{N m}^{-1}$ | **A1** |

spring constant = ................................................ $\text{N m}^{-1}$ [2]

**(ii)** the Young modulus of the material from which the sample is made.

| | |
|---|---|
| Young modulus or $E = \sigma / \varepsilon$ **or** $FL / Ax$ **or** $kL / A$ | **C1** |
| $E = (5.4 \times 3.2) / (4.0 \times 10^{-7} \times 0.06)$ **or** $90 \times 3.2 / (4.0 \times 10^{-7})$ | **C1** |
| $E = 7.2 \times 10^{8}$ Pa | **A1** |

Young modulus = ...................................................... Pa [3]

**(d)** Determine an estimate of the work done on the sample as it is extended from zero extension to its breaking point. Explain your reasoning.

| | |
|---|---|
| work done = area under graph | **B1** |
| = $(1.0 \pm 0.2)$ J | **A1** |

work done = ........................................................ J [2]

**(e)** A second sample of the same material has a larger cross-sectional area than the original sample but the same initial length. The two samples are each deformed with the limit of proportionality.

State and explain qualitatively how the spring constant of the second sample compares with that of the original sample.

| | |
|---|---|
| the extension will be smaller (for the same force on the thicker sample)<br>**or**<br>a greater force is required (to extend the thicker sample by the same amount)<br>**or**<br>spring constant is proportional to area | **M1** |
| the spring constant (of the second sample) will be greater | **A1** |

........

........

.. [2]

[Total: 12]

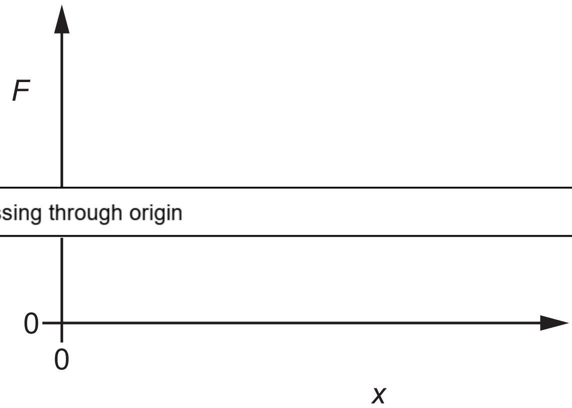
**4** **(a)** Define the Young modulus of a material.

| | |
|---|---|
| stress per unit strain | **B1** |

.................................................................................................................................... [1]

**(b)** A metal wire P that obeys Hooke's law is stretched within its limit of proportionality.

**(i)** On Fig. 4.1, sketch the variation of tensile force *F* in the wire with its extension *x*.




| | |
|---|---|
| straight line with positive gradient passing through origin | **B1** |

**Fig. 4.1**

[1]

**(ii)** State the name of the quantity represented by the gradient of the line in Fig. 4.1.

| | |
|---|---|
| spring constant | **B1** |

[1]

**(iii)** State the name of the quantity represented by the area under the line in Fig. 4.1.

| | |
|---|---|
| elastic potential energy (stored in wire) | **B1** |

[1]

**(c)** Another wire Q is made from a metal that has twice the Young modulus of the metal of wire P in **(b)**. Wire Q has the same volume as wire P but has double the cross-sectional area of wire P.

The two wires are extended by equal tensile forces within their limits of proportionality.

State and explain how the extension of wire Q compares with the extension of wire P.

| | |
|---|---|
| length (of Q) is half (the length of P) | **B1** |
| extension is proportional to length, and inversely proportional to area and Young modulus | **B1** |
| extension of Q is = ½ / (2 × 2)<br>= ⅛ times the extension of P | **B1** |

....................................................................................................................................

.................................................................................................................................... [3]

[Total: 7]

**3** **(a)** The variation of stress with strain for a metal P is shown in Fig. 3.1.

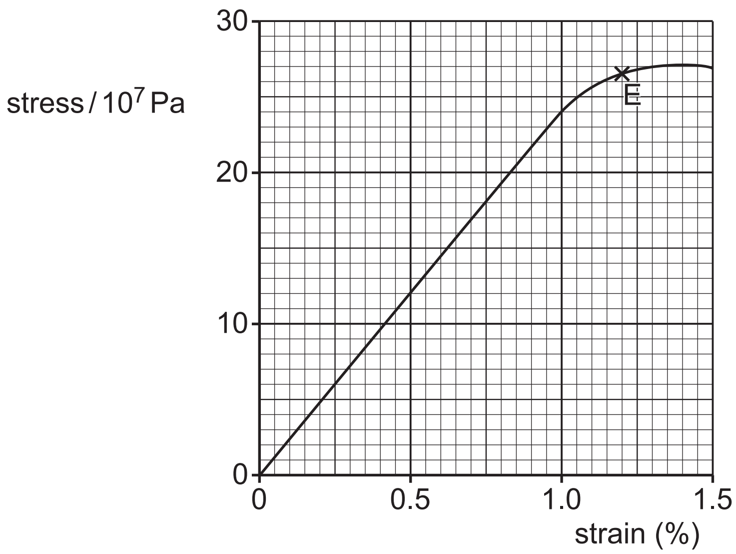



**Fig. 3.1**

Point E is the elastic limit of the metal.

**(i)** Use Fig. 3.1 to determine the Young modulus for P.

| | |
|---|---|
| $E = \sigma / \varepsilon$ **or** $E$ = gradient | **C1** |
| $E$ = e.g. $12 \times 10^7 / 0.0050$<br>$= 2.4 \times 10^{10}$ Pa | **A1** |

Young modulus = .................................................... Pa [2]

**(ii)** On the line in Fig. 3.1, draw a cross (×) to show the limit of proportionality.
Label this point Q. [1]

| | |
|---|---|
| cross drawn at (1.0%, $24 \times 10^7$ Pa), labelled Q | **B1** |

**(b)** State the conditions necessary for an object to be in equilibrium.

| | |
|---|---|
| resultant force (in any direction) is zero | **B1** |
| resultant moment / torque (about any point) is zero | **B1** |

[2]

**(c)** A wire is used to hold a uniform shelf AB horizontally in equilibrium as shown in Fig. 3.2.

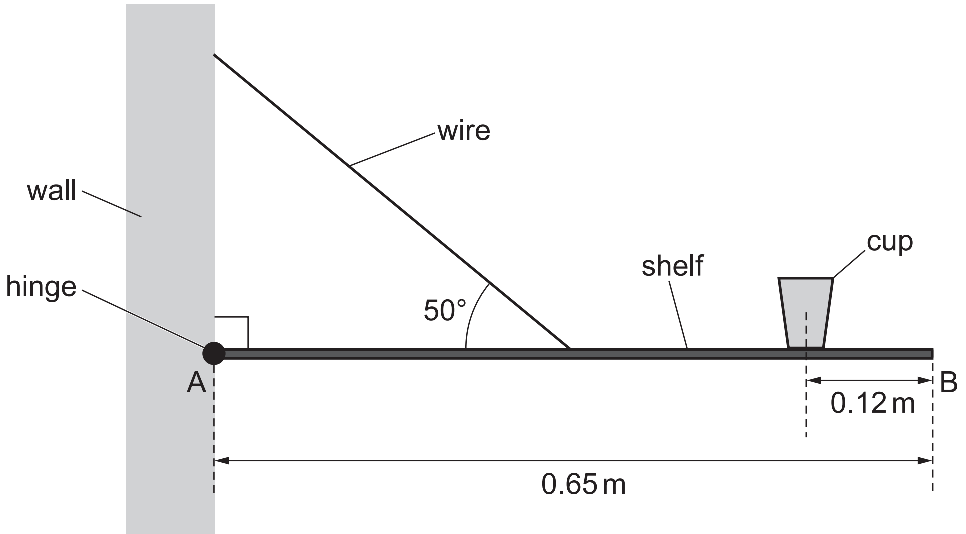



**Fig. 3.2** (not to scale)

The wire is connected to the midpoint of shelf AB at an angle of 50° to the horizontal. The shelf is attached to a wall by a hinge at A. The length of shelf AB is 0.65 m and its weight is 33 N.
A cup of weight 1.5 N rests on the shelf with its centre of gravity at a horizontal distance of 0.12 m from B.

**(i)** By taking moments about A, determine the tension in the wire.

| | |
|---|---|
| (moment =) $33 \times 0.65 / 2$<br>**or**<br>$1.5 \times (0.65 - 0.12)$<br>**or**<br>$T \sin 50° \times (0.65 / 2)$ | **C1** |
| sum of clockwise moments = sum of anticlockwise moments<br><br>$33 \times (0.65 / 2) + 1.5 \times (0.65 - 0.12) = T \sin 50° \times (0.65 / 2)$ | **C1** |
| tension = 46 N | **A1** |

tension = ...................................................... N [3]

**(ii)** The stress in the wire is $1.5 \times 10^7$ Pa.

Determine the radius of the wire.

| | |
|---|---|
| $\sigma = F / A$ | **C1** |
| $\pi r^2 = 46 / (1.5 \times 10^7)$<br>$r = 9.9 \times 10^{-4}$ m | **A1** |

radius = ........................................................ m [2]

**(iii)** More items are added to the shelf, doubling the stress in the wire. The wire is made of the metal P from **(a)**.

Use Fig. 3.1 to state and explain whether the wire will behave plastically or elastically as the stress doubles.

| | |
|---|---|
| elastic limit is not reached<br>**or**<br>(new) stress is less than (stress at) elastic limit<br>**or**<br>(new) strain is less than (strain at) elastic limit | **M1** |
| (so the wire behaves) elastically | **A1** |

..............................................................................................................................................

.............................................................................................................................................. [2]

[Total: 12]

**4** **(a)** Define:

**(i)** stress

| (normal) force per unit cross-sectional area | **B1** |
|---|---|

.............................................................................................................................. [1]

**(ii)** strain.

| extension per unit unstretched length | **B1** |
|---|---|

.............................................................................................................................. [1]

**(b)** Two wires X and Y, with equal unstretched lengths of 0.84 m, are suspended from fixed points that are at the same horizontal level. The lower ends of the wires are attached to a beam of negligible mass. The beam is horizontal and in equilibrium, as shown in Fig. 4.1.

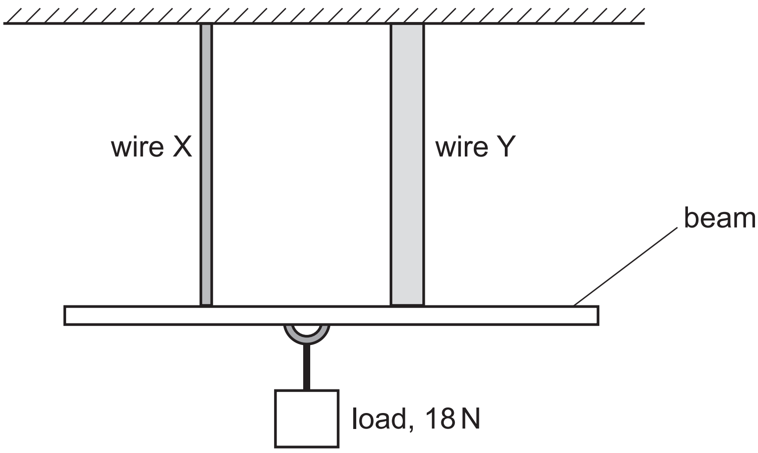



**Fig. 4.1**

Wire X is made from a metal that has a Young modulus of $1.9 \times 10^9$ Pa.
Wire Y is made from a different metal.

A load of weight 18 N is suspended from the beam at a point that is equidistant from the two wires. This load causes both wires to extend by 0.47 mm.

**(i)** Determine the cross-sectional area of wire X.

| | |
|---|---|
| $E = FL / Ax$ | **C1** |
| $A = (9.0 \times 0.84) / (1.9 \times 10^{9} \times 0.47 \times 10^{-3})$ | **C1** |
| $= 8.5 \times 10^{-6}\ \text{m}^2$ | **A1** |

cross-sectional area = .................................................... $\text{m}^2$ [3]

**(ii)** Wire Y has a greater diameter than wire X.

Explain, without calculation, whether the Young modulus of the metal from which wire Y is made is less than, the same as or greater than $1.9 \times 10^{9}$ Pa.

| | |
|---|---|
| *F*, *L* and *x* are all the same (for both wires / as in X)<br>**or**<br>*F* and strain are the same | **B1** |
| *A* is greater (for Y), so the Young modulus (for Y) is less than $1.9 \times 10^{9}$ Pa **or** less than that of wire X | **B1** |

........................................................................................................................ [2]

[Total: 7]